ENGINEERING PROGRAMMES

TEVET QUALIFICATIONS FRAMEWORK (TQF) - LEVEL 6

CURRICULUM CHART

FOR

DIPLOMA IN HEAVY EQUIPMENT ENGINEERING

CHART No. 352

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ACRONYMS

5S	Sort, Set in Order, Shine, Standardise and Sustain
AWD	All Wheel Drive
DLC	Data Link Connector
DTC	Diagnostic Trouble Codes
EFI	Electronic Fuel injection
EFI	Electronic Fuel Injection
HER	Heavy Equipment Repair
HOS	Head of Section
MIG	Metal Inert Gas
MIL	Malfunction Indicator Lamp
NORTEC	Northern Technical College
OWA	Oil and Water Analysis
PPC	Proportional Pressure Coolant
ST's	Special Service Tools
UNIDO	United Nations Industrial Development Organisation

1.0 ACKNOWLEDGEMENTS

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2.0 RATIONALE

Due to the rapid advancement of technology in infrastructure and extractive industries in recent years, a technologist needs to keep abreast with the essential knowledge to diagnose, service and repair heavy equipment successfully.

The modern day machineries are equipped with microprocessor controlled electronic systems installed on various mechanical and hydraulic components for the purpose of achieving precision controls and enhanced performance.

The role of microprocessor electronic control systems in modern machinery is beyond the reach of a conventional mechanic and current technician. These systems can only be diagnosed using specific scan tools which have the manufacturer's software to read and display the various fault codes.

This has led to a scenario change in the modern heavy equipment repair shops where mechanics are now getting replaced with technicians who are capable of operating multi-testers, scan tools, online data and use appropriate computer software to troubleshoot problems

With this rapid advancement in technology, there is a dire need to continuously revise and upgrade related curricula so that graduates are relevant to industry. This programme will therefore, provide the nation with qualified Heavy Equipment Engineering Technologists for employment in the formal and informal sectors with relevant competences to attend to modern equipment service needs.

3.0 PROGRAMME PURPOSE

To equip the trainees with knowledge, skills and appropriate attitudes to effectively and efficiently maintain and repair heavy equipment machines to manufacturer's standards.

4.0 PROGRAMME OUTCOMES

On completion of the programme the trainee will be able to;

4.1 Identify modern high speed diesel engines with variable valve lift mechanism and variable vane turbo charger
4.2 Repair Diesel electronic fuel injection system
4.3 Repair modern high speed diesel engines
4.4 Interpret and trace Electrical Wiring Diagram (Schematics) using multi-tester

4.5 Interpret hydraulic schematics and test circuits using pressure testing device
4.6 Service Automatic Transmission using special service tools (SST's)
4.7 Carry out preventive maintenance using appropriate service manual and special service tools
4.8 Read and interpret Diagnostic Trouble Codes (DTC's) using electronic scan tool

5.0 COURSE DURATION

The programme duration is fifteen (15) months or 1730 notional learning hours, inclusive of 480 hours industrial attachment.

6.0 COURSE OUTLINE

MODULE	TITLE	HOURS
352-01-A	Advanced Workshop Processes and Practice	150
352-02-A	Applied Hydraulics and Pneumatics	250
352-03-A	Applied Electricals I	150
352-04-A	Applied Electricals II	150
352-05-A	Electronic Fuel Injection- Diesel	150
352-06-A	Electronically Controlled Drive Train	150
352-07-A	Engine Maintenance	150
352-08-A	Maintenance Management Systems	100
	Industrial Attachment	480
	Total	**1,730**

7.0 TEACHING/LEARNING STRATEGIES

The teaching and learning strategies shall constitute the following:

3.1 Lecture method
3.2 Audio visual aids (Video conferencing, Charts, Pictures, Models)
3.3 Hand-outs
3.4 Demonstrations
3.5 Discussions
3.6 Simulations
3.7 Study tours
3.8 Practical training.

8.0 PROGRAMME EVALUATION

TEVETA shall evaluate the programme as follows:

8.1 Formative evaluation

4.1.1 **Purpose**: To determine an on-going basis whether the programme is being implemented as planned and to advise on improvements.

4.1.2 **Major areas of Evaluations:** Course Aims and Objectives student's entry requirements, course contents, teaching, learning activities, learning resources.

4.1.3 **Evaluation Instruments:** Questionnaires, structured interviews, observations, checklist, examinations/test records, participation and attendance in class.

4.1.4 **Sources of Information:** Students, trainees, administrative monitors and records.

4.2 Summative Evaluation

4.2.1 **Major areas of evaluation** - Course aims, trainees' entry requirements, course content, learning resources, teaching/learning activities, trainers assessors, examiners and graduates' performance in employment.

4.2.2 **Evaluation Instruments**:- Questionnaires, structured interviews, observations, checklists, records, final integrated examinations.

4.2.3 **Sources of Information**:- Trainees, trainers, assessors, examiners, administrators, sponsors, students dropouts, employers, union officials and the general public.

9.0 ENTRY REQUIREMENTS:

9.1 Direct Entry

There shall be no direct entry to program.

9.2 Mature age entry

A minimum of Advanced Certificate in Heavy Equipment Repair from an accredited institution and a minimum of two (2) years working experience in relevant industry.

10.0 STUDENT ASSESSMENT

Trainees shall be subjected to end of year final examinations set by TEVETA.

10.1	Continuous assessments	40%
10.2	Final Examinations	60%
10.3	Pass Mark per Subject	50%

11.0 ATTENDANCE

The candidates must have an attendance of minimum 85% to be eligible for the final examinations

12.0 PROGRESSION REQUIREMENTS

12.1 Students who fail one (1) or two (2) subjects shall be allowed to progress and will be allowed two (2) sittings.

13.0 STAFF QUALIFICATION

13.1 Minimum of Bachelors in Mechanical/Automobile Engineering and three years industrial experience.

13.2 Diploma in Heavy Equipment Repair or equivalent and at least five (5) Years of Industrial experience in a reputable organisation.

13.3 Teaching Qualification

13.4 Accredited Trainer

14.0 CERTIFICATION

Successful candidates will be awarded a Diploma in Heavy Equipment Engineering by The Technical Education Vocational and Entrepreneurship Training Authority (TEVETA)

MODULE 352-01-A ADVANCED WORKSHOP PROCESSES AND PRACTICE

NOMINAL DURATION: 150 HOURS CREDITS: 15

MODULE PURPOSE: To equip trainees with knowledge, skills and appropriate attitudes to effectively and efficiently diagnose faults and carry out repairs in a heavy duty workshop.

LEARNING OUTCOMES: On completion of this module, trainees will be able to:

1. Organise work space
2. Demonstrate safe working practices
3. Use fasteners
4. Use lifting devises
5. Analyse fluid reports
6. Perform basic welding
7. Simulate machine operations

UNIT A1.1 ORGANISING WORK SPACE

1.1.1 Elements of 5S
1.1.2 Organise physical workspace for functionality and productivity
1.1.3 Workplace and Physical barriers

UNIT A1.2 DEMONSTRATING SAFE WORKING PRACTICES

1.2.1 Occupational Health and safety hazards
1.2.2 Hazardous workplace and materials
1.2.3 Personal protective equipment
1.2.4 Workplace-related legislation

UNIT A1.3 USING FASTENERS

1.3.1 Types of fasteners
1.3.2 Strength of fasteners
1.3.3 Application of fasteners
1.3.4 Torqueing techniques

UNIT A1.4 USING LIFTING DEVICES

1.4.1 Safety and certification requirements
1.4.2 Sling work
1.4.3 Wire rope

1.4.4 Chain blocks

UNIT A1.5 ANALYSING FLUID REPORTS

1.5.1 Methods of carrying out OWA
1.5.2 Item analysis
1.5.3 Failure forecast

UNIT A1.6 PERFORMING WELDING

1.6.1 Fundamentals of welding
1.6.2 Types of welding
1.6.3 Application of welding

UNIT A1.7 SIMULATING MACHINE OPERATIONS

1.7.1 Familiarization to heavy equipment machinery
1.7.2 Simulating of wheel machinery
1.7.3 Simulating of chain machinery

MODULE NO: 352-02-A APPLIED HYDRAULICS AND PNEUMATICS

NOMINAL DURATION: 250 HOURS CREDITS: 25

MODULE PURPOSE: To equip trainees with knowledge, skills and appropriate attitudes to effectively and efficiently maintain heavy duty hydraulics and pneumatics systems.

LEARNING OUTCOMES: On completion of this module, the trainees will be able to:

1. Interpret hydraulic and pneumatic schematic diagram
2. Apply the working principles of brake and steering circuits used in heavy equipment
3. Diagnose hydraulic and pneumatic system
4. Apply the principle of electro-hydraulic suspension system.
5. Apply the principles of electro-pneumatic suspension system
6. Apply the principles of electro-pneumatic, hydro-pneumatic and electro-hydraulic braking system
7. Identify the various pneumatic and hydraulic accessories systems
8. Design pneumatic and hydraulic circuit

UNIT A2.1 INTERPRETING HYDRAULIC AND PNEUMATIC SCHEMATIC DIAGRAMS

2.1.1 Hydraulic and pneumatic components
2.1.2 Hydraulic and Pneumatic symbols
2.1.3 Schematic diagram

UNIT A2.2 APPLYING THE WORKING PRINCIPLES OF BRAKE AND STEERING CIRCUITS USED IN HEAVY EQUIPMENT

2.2.1 Construction and operation of hydrostatic steering system
2.2.2 Brake systems
2.2.3 Steering systems

UNIT A2.3 DIAGNOSING HYDRAULIC AND PNEUMATIC SYSTEM

2.3.1 Testing equipment
2.3.2 Troubleshooting the system
2.3.3 Final inspection

UNIT A2.4 APPLYING THE PRINCIPLE OF ELECTRO-HYDRAULIC SUSPENSION SYSTEM

2.4.1 Safety and preparation
2.4.2 Removal and refitting of electro-hydraulic component
2.4.3 Final inspection

UNIT A2.5 APPLYING THE PRINCIPLES OF ELECTRO-PNEUMATIC SUSPENSION SYSTEM

2.5.1 Safety and preparation
2.5.2 Removal and refitting of electro-pneumatics component
2.5.3 Final inspection

UNIT A2.6 APPLYING THE PRINCIPLES OF ELECTRO-PNEUMATIC, HYDRO-PNEUMATIC AND ELECTRO-HYDRAULIC BRAKING SYSTEM

2.6.1 Construction and operation
2.6.2 Components
2.6.3 Inspection and service

UNIT A2.7 IDENTIFYING THE VARIOUS PNEUMATIC AND HYDRAULIC ACCESSORIES SYSTEMS

2.7.1 Pneumatic & hydraulic systems
2.7.2 Components
2.7.3 Inspection and service

UNIT A2.8 DESIGNING SIMPLE PNEUMATIC AND HYDRAULIC CIRCUIT

2.8.1 Introduction to circuits designs
2.8.2 Hydraulic circuits
2.8.3 Pneumatics circuits

MODULE NO: 352-03-A APPLIED ELECTRICALS I

NOMINAL DURATION: 150 HOURS CREDITS: 15

MODULE PURPOSE: To equip trainees with knowledge, skills and appropriate attitudes to effectively and efficiently repair electrical systems.

LEARNING OUTCOMES: On completion of this module, the trainees will be able to:

1. Perform electrical circuit tests
2. Modify electrical circuits
3. Apply the principles of electrical machines - capacitors
4. Inspect rectification circuits - inductors
5. Apply the working principles of electronic devices

UNIT A3.1 PERFORMING ELECTRICAL CIRCUIT TESTS

3.1.1 Types of measuring instruments
3.1.2 Measuring voltage
3.1.3 Measuring current
3.1.4 Measuring resistance
3.1.5 Measuring waveform

UNIT A3.2 MODIFYING ELECTRIC CIRCUITS

3.2.1 Series circuit
3.2.2 Parallel circuit
3.2.3 Load on a circuit
3.2.4 Wire specifications

UNIT A3.3 APPLYING THE PRINCIPLES OF ELECTRICAL MACHINES

3.3.1 Motors
3.3.2 Generators
3.3.3 Alternators
3.3.4 Capacitors

UNIT A3.4 INSPECTING RECTIFICATION CIRCUIT

3.4.1 Diode
3.4.2 Half-wave rectification circuit
3.4.3 Full-wave rectification circuit

UNIT A3.5 APPLYING THE WORKING PRINCIPLES OF ELECTRONIC DEVICES

3.5.1 Transistors
3.5.2 Thermistor
3.5.3 Piezo element
3.5.4 Integrated circuits
3.5.5 Logic gates

MODULE NO: 352-04-A APPLIED ELECTRICALS II

NOMINAL DURATION: 150 HOURS CREDITS: 15

MODULE PURPOSE: To equip trainees with knowledge, skills and appropriate attitudes to effectively and efficiently diagnose and repair electricals.

LEARNING OUTCOMES: On completion of the module, the trainees will be able to:

1. Analyse the layout of electrical systems
2. Troubleshoot circuits
3. Troubleshoot ignition system
4. Troubleshoot starting system
5. Troubleshoot service charging system
6. Configure programmable Logic Controllers

UNIT A4.1 ANALYSING THE LAYOUT OF ELECTRICAL SYSTEMS

4.1.1 Electrical wiring diagram
4.1.2 Wiring harness layout
4.1.3 Component identification

UNIT A4.2 TROUBLESHOOTING ELECTRICAL CIRCUITS

4.2.1 Series circuit
4.2.2 Parallel circuit
4.2.3 Series/Parallel circuit

UNIT A4.3 TROUBLESHOOTING IGNITION CIRCUIT

4.3.1 Power flow
4.3.2 Component inspection
4.3.3 Circuit faults

UNIT A4.4 TROUBLESHOOTING STARTING SYSTEM

5.4.1 Power flow
5.4.2 Components inspection
5.4.3 Circuit faults

UNIT A4.5 TROUBLESHOOTING CHARGING SYSTEM

4.5.1 Power flow
4.5.2 Components inspection

4.5.3 Circuit faults

UNIT A4.6 CONFIGURING PROGRAMABLE LOGIC CONTROLLER

4.5.1 Programmable logic controller
4.5.2 Maintenance
4.5.3 Programming

MODULE NO: 352-05-A ELECTRONIC FUEL INJECTION - DIESEL

NOMINAL DURATION: 150 HOURS CREDITS: 15

MODULE PURPOSE: To equip trainees with knowledge, skills and appropriate attitudes to effectively and efficiently diagnose engine faults.

LEARNING OUTCOMES: On completion of the module, trainees will be able to:

1. Explain the fundamentals of electronic diesel fuel injection system
2. Apply the principles of operation of electronic diesel fuel injection systems components
3. Interpret electrical schematic wiring diagram in the engine
4. Inspect electronic diesel fuel injection systems using diagnostic tooling
5. Replace electronic diesel fuel injection systems components on an engine
6. Explain the fundamentals of multiplex network systems used in transport sector.

UNIT A5.1 EXPLAINING THE FUNDAMENTALS OF ELECTRONIC DIESEL FUEL INJECTION SYSTEM

5.1.1 Purpose
5.1.2 System outline
5.1.3 Types of Electronic Fuel Injection

UNIT A5.2 APPLYING THE PRINCIPLES OF OPERATION OF ELECTRONIC DIESEL FUEL INJECTION SYSTEMS COMPONENTS

5.2.1 Status switches and Sensors
5.2.2 Actuators and fuel pumps
5.2.3 Glow plug control algorithm
5.2.4 Malfunction indicator lamp (MIL)
5.2.5 Electronic control module

UNIT A5.3 INTERPRETING ELECTRICAL CIRCUITS ON ACTUAL MACHINE

5.3.1 Circuit diagram
5.3.2 Inspection
5.3.3 Troubleshooting

UNIT A5.4 INSPECTING ELECTRONIC DIESEL FUEL INJECTION SYSTEMS USING DIAGNOSTIC TOOLING

5.4.1 Operation of a scan tool
5.4.2 Data link connector (DLC) pin configuration
5.4.3 Connecting diagnostic tool
5.4.4 Troubleshooting diagnostic trouble codes (DTC)

UNIT A5.5 REPLACING ELECTRONIC DIESEL FUEL INJECTION SYSTEMS COMPONENTS ON AN ENGINE

5.5.1 Circuit diagram
5.5.2 Removal of components
5.5.3 Refitting the components

UNIT A5.6 EXPLAINING THE FUNDAMENTALS OF MULTIPLEX NETWORK SYSTEMS USED IN TRANSPORT SECTOR

5.6.1 System Operation
5.6.2 Components
5.6.3 Types
5.6.4 Diagnostics capabilities

MODULE NO: 352-06-A ELECTRONICALLY CONTROLLED DRIVE TRAIN

NOMINAL DURATION: 150 HOURS CREDITS 150

MODULE PURPOSE: To equip trainees with knowledge, skills and appropriate attitudes to effectively and efficiently troubleshoot drive train assemblies and components.

LEARNING OUTCOMES: On completion of the module, the trainees will be able to:

1. Explain the operating principles of an electronically controlled drive train
2. Troubleshoot problem on electronically controlled drivetrain according to manufacturer standards
3. Maintain electronically controlled transmission according to manufacturer specification
4. Explain the operating principles of an electronically controlled transfer case
5. Maintain electronically controlled transfer case according to manufacturer specification
6. Describe the operation of All Wheel Drive transfer cases
7. Maintain All Wheel Drive transfer case according to manufacturer specification

UNIT A6.1 EXPLAINING THE OPERATING PRINCIPLES OF AN ELECTRONICALLY CONTROLLED DRIVE TRAIN

6.1.1 System outline and types
6.1.2 Sensors and Actuators
6.1.3 Transmission control module
6.1.4 Valve body assembly
6.1.5 Torque convertors lockups

UNIT A6.2 TROUBLESHOOTING ELECTRONICALLY CONTROLLED DRIVETRAIN ACCORDING TO MANUFACTURER STANDARDS

6.2.1 Troubleshooting
6.2.2 Tools / Instruments
6.2.3 Tests
6.2.4 Inspection techniques

UNIT A6.3 MAINTAINING ELECTRONICALLY CONTROLLED TRANSMISSION ACCORDING TO MANUFACTURER SPECIFICATION

6.3.1 Publications
6.3.2 Disassembly
6.3.3 Inspection
6.3.4 Reassembly

UNIT A6.4 EXPLAINING THE OPERATING PRINCIPLES OF AN ELECTRONICALLY CONTROLLED TRANSFER CASE

6.4.1 System outline and types
6.4.2 Mechanical components function
6.4.3 Sensors and actuators
6.4.4 Transfer case control module
6.4.5 Shifting mechanism

UNIT A6.5 MAINTAINING AND REPAIR ELECTRONICALLY CONTROLLED TRANSFER CASE ACCORDING TO MANUFACTURER RECOMENDATION

6.5.1 Publication
6.5.2 Disassembly
6.5.3 Inspection
6.5.4 Reassembly

UNIT A6.6 DESCRIBING THE OPERATION OF ALL WHEEL DRIVE TRANSFER CASES

6.6.1 System outline and types
6.6.2 Components function
6.6.3 Differential lock
6.6.4 High and low range shifting mechanism

UNIT A6.7 MAINTAINING ALL WHEEL DRIVE TRANSFER CASES ACCORDING TO MANUFACTURER SPECIFICATION

6.7.1 Publication
6.7.2 Disassemble
6.7.3 Inspection
6.7.4 Reassembly

MODULE NO: 352-07-A ENGINE MAINTENANCE

NOMINAL DURATION: 150 HOURS CREDITS: 15

MODULE PURPOSE: To equip trainees with knowledge, skills and appropriate attitudes to effectively and efficiently safely carry out repairs on Engine

LEARNING OUTCOMES: On completion of the module, trainees will be able to:

1. Disassemble the heavy duty diesel engine using special tools and follow recommended procedure
2. Inspect engine parts and assemblies using recommended procedure
3. Reassemble the heavy duty diesel engine using proper special tools and recommended procedure
4. Performance test using a dynamometer
5. Apply technologies in diesel engine

UNIT A7.1 DISASSEMBLING THE MODERN HIGH SPEED DIESEL ENGINE USING SPECIAL TOOLS AND RECOMMENDED PROCEDURE

7.1.1 Preparation – Tools and place
7.1.2 Disassemble
7.1.3 Cleaning

UNIT A7.2 INSPECTING ENGINE PARTS AND ASSEMBLIES USING RECOMMENDED PROCEDURE

7.2.1 Preparation
7.2.2 Inspection
7.2.3 Part number reconciliation

UNIT A7.3 REASSEMBLING HEAVY DUTY DIESEL ENGINE USING SPECIAL TOOLS AND RECOMMENDED PROCEDURE

7.3.1 Preparation
7.3.2 Reassembling
7.3.3 Overhaul report

UNIT A7.4 PERFORMING TEST USING A DYNAMOMETER

7.4.1 Preparation
7.4.2 Inspection
7.4.3 Testing
7.4.4 Analysis report and corrective actions

UNIT A7.5 APPLYING TECHNOLOGIES IN DIESEL ENGINES

7.5.1 Variable valve timing technology
7.5.2 Variable vane turbo charger
7.5.3 Intake shutter valves
7.5.4 Electronically controlled Exhaust Gas Recirculation

MODULE NO.352-08-A MAINTENANCE SYSTEMS OF HEAVY EQUIPMENT ENGINEERING

NOMINAL DURATION: 100 HOURS CREDITS: 15

MODULE PURPOSE: To equip trainees with knowledge, skills and appropriate attitudes to effectively and efficiently maintain Heavy Equipment Machines.

LEARNING OUTCOMES: On completion of the module, trainees will be able to:

1. Demonstrate skills on the various types of maintenance
2. Perform maintenance on highway truck in line with manufacturer recommendations
3. Perform maintenance on Excavator in-line with manufacturer recommendations
4. Perform maintenance on loader in line with manufacturer recommendations
5. Perform maintenance on dump truck in line with manufacturer recommendations

UNIT A8.1 DESCRIBING VARIOUS TYPES OF MAINTENANCE

8.1.1 Preventive maintenance
8.1.2 Predictive maintenance
8.1.3 Corrective maintenance
8.1.4 Risk based maintenance
8.1.5 Scheduled maintenance

UNIT A8.2 PERFORMING MAINTENANCE ON HIGHWAY TRUCK IN LINE WITH MANUFACTURER RECOMENDATIONS

8.1.1 Maintenance chart
8.1.2 maintenance chart operation
8.1.3 Signing off the maintenance sheet

UNIT A8.3 PERFORMING MAINTENANCE ON EXCAVATOR IN-LINE WITH MANUFACTURER RECOMENDATIONS

8.3.1 Maintenance chart
8.3.2 Performing maintenance chart operation
8.3.3 Signing off the maintenance sheet

UNIT A8.4 PERFORMING MAINTENANCE ON LOADER IN-LINE WITH MANUFACTURER RECOMENDATIONS

8.4.1 Maintenance chart
8.4.2 Performing maintenance chart operation
8.4.3 Signing off the maintenance sheet

UNIT A8.5 PERFORMING MAINTENANCE ON DUMP TRUCK IN LINE WITH MANUFACTURER SPECIFICATION RECOMENDATIONS

8.5.1 Maintenance chart

8.5.2 Performing maintenance chart operation

8.5.3 Signing off the maintenance sheet

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